

**System-Level Design and Simulation of
a Silicon-Photonic NAND Gate Using
an All-Optical Artificial Neural
Network, with a Cascaded XOR Study**

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I. Introduction

Artificial neural networks and photonic devices are both major areas of interest because they offer the possibility of fast, trainable, and highly parallel information processing. Although artificial neural networks and machine learning are now mature technologies, they have been implemented mainly on electronic hardware. At the same time, the growing demand for high-speed data processing and communication continues to highlight the bandwidth and speed limitations of purely electronic signals. For this reason, it is important to investigate whether neural-network operations can be realized using optical devices.

In an artificial neural network, the linear unit is primarily the matrix multiplication operation, and this can be implemented in the optical domain in several ways. One common approach uses optical interference in Mach–Zehnder interferometers, while other photonic matrix-processing architectures can also be used, including directional-coupler-based structures such as those demonstrated by Feldmann et al. [1]. The nonlinear activation function, however, has generally been explored in two forms. The first is purely all-optical activation, in which the signal remains in the optical domain through the nonlinear operation. The second is electro-optic activation, in which part or all of the optical signal is converted into an electrical signal, the nonlinear function is applied, and the result is then used to modulate or regenerate the optical output. Both approaches are actively studied, although they are not equally mature. In principle, a fully all-optical neural network could provide lower latency and lower energy consumption.

Shi et al. used a germanium-silicon photodiode in which nonlinear activation and monitoring are combined in the same device [2]. In their structure, part of the Ge region has no electrode. In that electrode-free region, photogenerated carriers accumulate instead of being swept out quickly. This carrier accumulation increases free-carrier absorption (FCA) in the Ge film, which makes the output optical power depend nonlinearly on the input optical power. At the same time, the collected carriers provide an electrical monitoring signal. Therefore, the device enables all-optical nonlinear activation together with non-intrusive monitoring in the same structure.

This work investigates, at the system level, whether a silicon-photonic NAND gate built from a linear optical unit and an optical nonlinear activation block can be realized and then further studied by cascading the NAND gate to examine XOR realization. One advantage of the NAND gate is that the NAND function is linearly separable, unlike XOR and XNOR. Another advantage is that the NAND gate can be cascaded to realize all other logic gates, as Floyd explained in his textbook [3]. Spears used the XOR gate as a neural-network example and showed that a linear model fails, while a small network with a ReLU activation function can realize XOR exactly [4], as shown in Eqn. 1.

This work is a system-level numerical study and does not include device fabrication or experimental validation.

$$y = \max(0, x_1 + x_2) - 2 \max(0, x_1 + x_2 - 1) \quad (1)$$

II. NAND gate

a. Architecture

The optical linear unit in this work consists of a single MZI formed by 50:50 directional couplers and trainable internal and external phase shifts, which act as the trainable parameters of the optical neuron. A bias term is also included at the system level, as shown in Fig. 1. The nonlinear activation function is then applied in the optical domain. At the ANN abstraction level, this structure is treated as a single-neuron optical block. This is sufficient for NAND because NAND is linearly separable, whereas XOR is not, and therefore generally requires a deeper or wider network representation.

The proposed architecture is inspired by the work of Shen et al., who used programmable Mach–Zehnder interferometers as the linear optical units of a coherent nanophotonic neural network [5]. In their implementation, multiple optical interference units and phase shifters were trained to perform a classification task. In the present work, the same general idea is reduced to a much smaller system-level block in order to study a single logical function, namely NAND, before examining whether this trained block can be cascaded to study XOR realization.

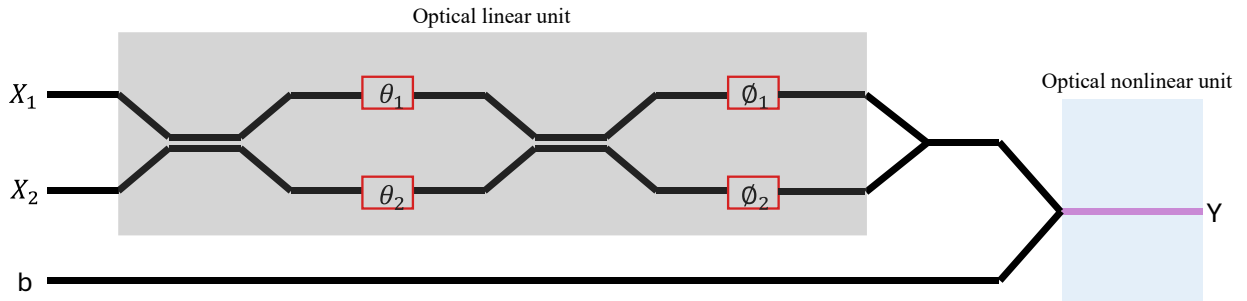


Figure 1: Schematic of the optical neural network to realize a NAND gate

b. Nonlinear Activation Function

The optical nonlinear function used in this work is implemented using saturable absorbers, where the transmission increases nonlinearly with the input light intensity following Eqn. 2, where σ is the absorption cross section, τ_s is the radiative lifetime of the absorber material, T_o is the initial transmittance which depends on the design of saturable absorbers, I_{in} is the incident intensity, and

T_m is the transmittance of the absorber. In this way, the output intensity varies with the input intensity in a nonlinear manner as in Eqn. 3 and Fig. 2.

In this work, $T_0 = 0.7$ is used in the training model, while $T_0 = 0.1$ is used only in Figures 2 and 3 to visually emphasize the nonlinear response, following the form shown in the supplementary discussion of Shen et al. As shown in Fig. 3, the field-domain response is an odd and continuously increasing nonlinear function of the input field amplitude. For logic-gate applications, however, a more threshold-like or saturating nonlinear response would generally be more suitable.

$$\sigma \tau_s I_{in} = \frac{1}{2} \frac{\ln(T_m/T_0)}{1 - T_m} \quad (2)$$

$$I_{out} = I_{in} * T_m(I_{in}) \quad (3)$$

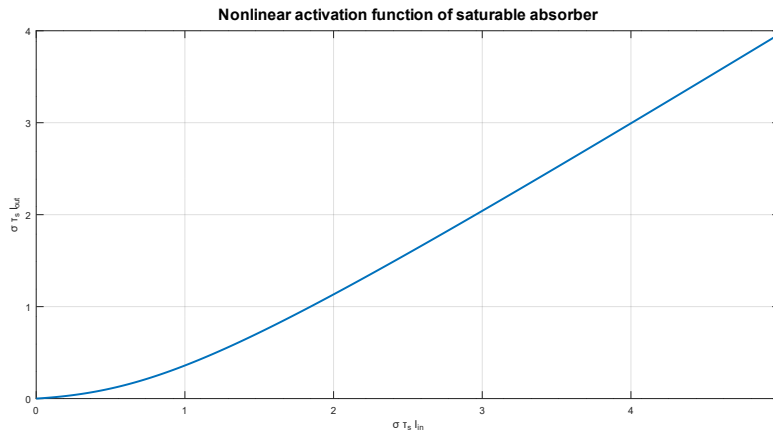


Figure 2: Normalized output intensity vs normalized input intensity

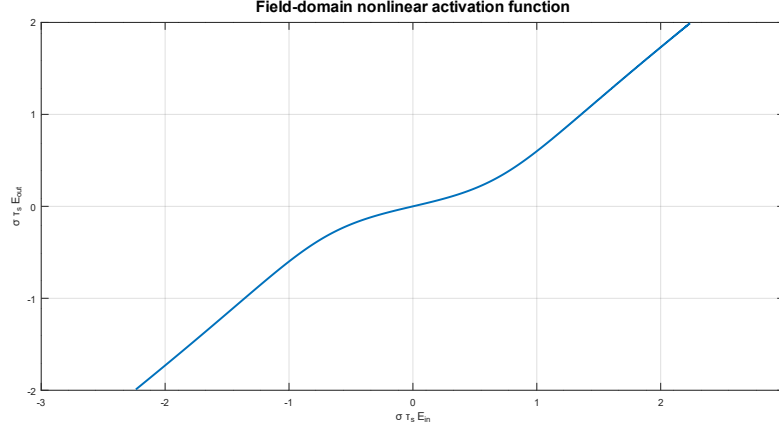


Figure 3: Normalized output field vs normalized input field

c. Network transfer matrix

This section presents the transfer function of the optical linear unit followed by the nonlinear activation function, with the unknown values $\theta_1, \theta_2, \phi_1, \phi_2$, and b treated as trainable parameters. These parameters are then supplied to the Python optimization model in order to minimize the loss with respect to the NAND truth table.

$$\begin{aligned} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \begin{bmatrix} e^{-j\theta_1} & 0 \\ 0 & e^{-j\theta_2} \end{bmatrix} \begin{bmatrix} 1 & -j \\ -j & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \\ \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} e^{-j\theta_1} - e^{-j\theta_2} & -j(e^{-j\theta_1} + e^{-j\theta_2}) \\ -j(e^{-j\theta_1} + e^{-j\theta_2}) & e^{-j\theta_2} - e^{-j\theta_1} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \\ \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} &= \begin{bmatrix} e^{-j\phi_1} & 0 \\ 0 & e^{-j\phi_2} \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} \\ Z &= Z_1 + Z_2 + b \\ Y &= f(Z) \end{aligned}$$

The Python implementation available in the accompanying GitHub repository performs a system-level optimization of the proposed optical NAND gate by modeling the device as a 2×2 linear interferometric block followed by a nonlinear saturable-absorber activation. For each input pair $(0,0)$, $(0,1)$, $(1,0)$, and $(1,1)$, the input field vector is first passed through the 3 dB coupler matrix, the internal phase-shift matrix parameterized by θ_1 and θ_2 , and the external phase-shift matrix parameterized by ϕ_1 and ϕ_2 . The two resulting optical fields are then coherently summed, and a

bias field b is added to form the pre-activation field z . This field is applied to the nonlinear block, where the transmission T_m is found numerically from the Shen saturable-absorber relation, and the output field is then computed. In the present implementation, the training model uses $T_0 = 0.7$, which makes the absorber relatively transparent at low input power and therefore softer than the earlier illustrative $T_0 = 0.1$ case used only in the appendix plots. The output intensity $|Y|^2$ is then evaluated for all four NAND states and normalized so that the response can be compared directly with the NAND target vector $[1, 1, 1, 0]$.

To determine the optimal device settings, the code defines a mean-squared-error loss between the normalized simulated outputs and the desired NAND truth table, together with a small regularization term on the bias. The trainable parameters are optimized using a multi-start strategy based on `scipy.optimize.minimize` with the Nelder–Mead method, so that the search is repeated from several random initial points and the best result is retained. After optimization, the phase values are wrapped into the interval $[0, 2\pi)$, and the bias is snapped to a practical discrete optical field value, for example, in increments of 0.5, so that a numerically obtained value such as 0.97 can be replaced by the more practical value 1.0. The final section of the code then re-evaluates the optical NAND gate using these practical parameters and reports the linear pre-activation field, the activated optical field, and the normalized output intensities.

III. Results

The optimized parameters obtained from the Python-based training are

$$\begin{aligned}\theta_1 &= 5.514948, \theta_2 = 2.373355 \\ \phi_1 &= 4.942147, \phi_2 = 6.019105 \\ b_{\text{trained}} &= 0.974835, b_{\text{practical}} = 1.000000\end{aligned}$$

The best loss function value before snapping the bias to a practical value is

$$\mathcal{L}_{\min} = 9.621046529538 \times 10^{-5}$$

Using the practical value $b = 1$, the linear field before the nonlinear activation becomes

$$\begin{aligned}Z(0,0) &= 1.000000 + j0.000000 \\ Z(0,1) &= 0.487169 - j0.858490 \\ Z(1,0) &= 0.487169 + j0.858490 \\ Z(1,1) &= -0.025661 + j0.000000\end{aligned}$$

After applying the saturable-absorber nonlinear activation, the output field is

$$\begin{aligned} Y(0,0) &= 0.939979 + j0.000000 \\ Y(0,1) &= 0.457426 - j0.806076 \\ Y(1,0) &= 0.457426 + j0.806076 \\ Y(1,1) &= -0.021474 + j0.000000 \end{aligned}$$

The corresponding normalized output intensities used for comparison with the NAND truth table are

$$\begin{aligned} \tilde{y}(0,0) &= 1.000000 \\ \tilde{y}(0,1) &= 0.972200 \\ \tilde{y}(1,0) &= 0.972200 \\ \tilde{y}(1,1) &= 0.000522 \end{aligned}$$

Using a decision threshold of 0.5 on the normalized output intensity, the resulting hard decision is $(0,0) \rightarrow 1$, $(0,1) \rightarrow 1$, $(1,0) \rightarrow 1$, and $(1,1) \rightarrow 0$, which matches the desired NAND truth table. Therefore, at the system level, the optimized single optical block successfully reproduces NAND behavior under the adopted training.

IV. XOR gate

In this section, the trained photonic NAND block is cascaded in the same logical arrangement commonly used in electronic implementations of XOR, as shown in Fig. 4. Three cases are examined. In all three cases, the same trained optical NAND block is reused as the basic element. Thus, the study here is not about retraining a new XOR network, but about testing whether the previously trained NAND block remains useful when it is cascaded.

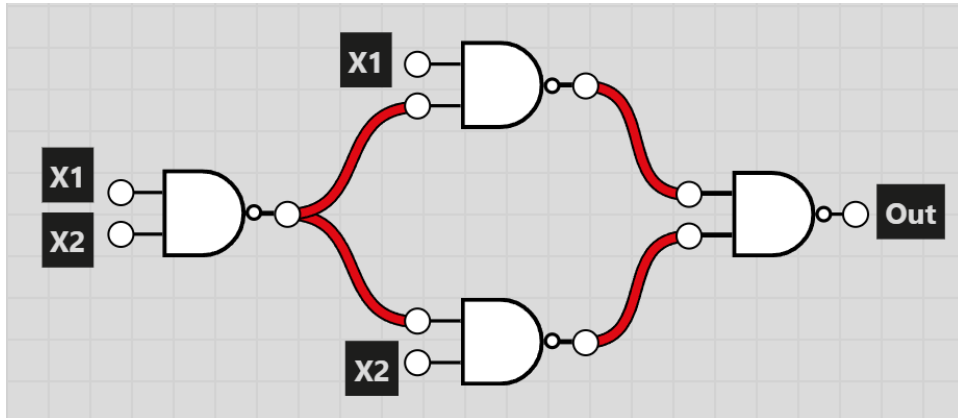


Figure 4: Cascading NAND into an XOR gate

a. Case 1: XOR realization with threshold restoration

In the first case, a restoration step is applied after each NAND stage. The raw output intensity of the NAND block is first normalized by a reference logic-1 intensity I_{ref} ,

$$I_{\text{norm}} = \frac{I_{\text{NAND}}}{I_{\text{ref}}}$$

and then converted into a logic value through thresholding:

$$x_{\text{logic}} = \begin{cases} 1, & I_{\text{norm}} > 0.5, \\ 0, & I_{\text{norm}} \leq 0.5. \end{cases}$$

The XOR gate is then formed from four restored NAND stages:

$$P = \text{NAND}(A, B)$$

$$Q = \text{NAND}(A, P)$$

$$R = \text{NAND}(B, P)$$

$$\text{XOR} = \text{NAND}(Q, R)$$

This case gives the exact XOR truth table because each intermediate stage is forced back to an ideal binary level before being used by the next stage. However, this restoration process is not a purely optical operation. It is best interpreted as a system-level regeneration model, corresponding conceptually to an optical-to-electrical-to-optical decision step. Therefore, Case 1 is useful for verifying the logical correctness of the cascade, but it does not represent a fully all-optical XOR implementation.

b. Case 2: XOR realization with gain saturation

In the second case, the hard threshold is replaced by a soft restoring stage based on gain with saturation. After the saturable-absorber output of each NAND block is obtained, the field is passed through a gain-saturation model of the form shown in Eqn. 4.

$$y_{\text{GS}} = \sqrt{\frac{G_0}{1 + |y_{\text{SA}}|^2 / I_{\text{sat}}}} y_{\text{SA}} \quad (4)$$

where G_0 is the small-signal gain and I_{sat} is the saturation intensity. This model amplifies weak outputs while compressing stronger outputs toward a limited range. It is therefore more compatible

with optical implementation than a hard digital threshold. A practical physical example of such a behavior is a semiconductor optical amplifier (SOA) or an erbium-doped waveguide amplifier, both of which exhibit gain saturation at high optical powers. This gain-with-saturation relation is used here as a phenomenological system-level restoration model. It is not taken directly from the Shen paper, but it is consistent with the idea of introducing an optical amplification stage after the nonlinear unit.

This case did not produce the correct XOR truth table. Although gain saturation improved signal survival, it did not restore the outputs to the required Boolean levels. As a result, the later NAND stages still operated on analog optical fields rather than ideal binary values, and the XOR output was not correct.

c. Case 3: XOR realization without restoration

In the third case, the XOR gate is formed by directly cascading the optical NAND blocks with no restoration stage or gain block between them. In this case, the output of each NAND block is fed forward as a raw complex optical field. Since the later stages no longer receive exact binary inputs, the Boolean NAND composition is not preserved exactly. In addition, the saturable absorber introduces nonlinearity but remains lossy, so the signal is not naturally regenerated through the cascade. For this reason, the final XOR output generally deviates from the ideal XOR truth table. This case, therefore, serves as the fully analog optical baseline and shows that nonlinearity alone is not sufficient for robust cascaded logic in the present architecture.

Overall, only Case 1 reproduced the XOR truth table, and it did so by relying on stage-by-stage restoration rather than a purely optical cascade. Cases 2 and 3 did not reproduce the exact XOR truth table under the present saturable-absorber-based NAND model, as shown in Fig. 5. This indicates that, in the present architecture, direct reuse of the trained NAND block is not sufficient for exact fully optical XOR realization.

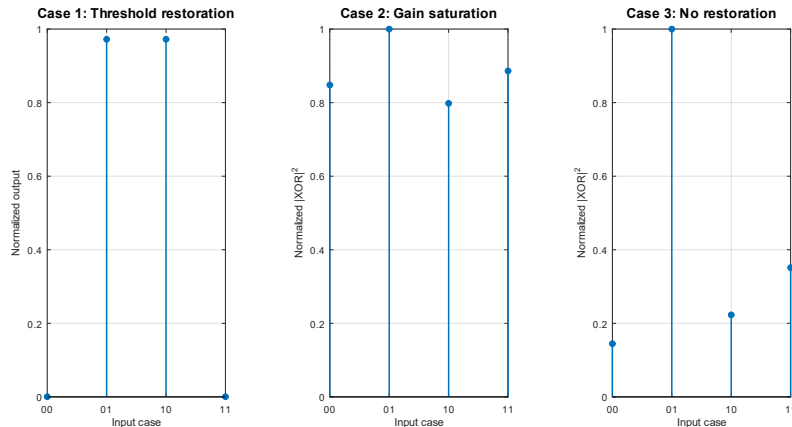


Figure 5: Output of the XOR gate in each case

V. Conclusion

The results of this work show that restoration is the key issue in cascaded optical logic. In Case 1, the XOR truth table was obtained because each NAND stage was followed by threshold restoration, which reset the intermediate outputs to ideal binary levels. However, this case should be interpreted only as a system-level reference, since it effectively corresponds to regeneration rather than a fully all-optical implementation. In Case 2, gain saturation improved signal survival in a more optical-compatible way, but it still did not restore the outputs to the required Boolean levels, so the XOR output was not correct. In Case 3, direct analog cascading without restoration caused the intermediate signals to remain continuous optical fields, and the final output also failed to reproduce the target XOR truth table. Therefore, within the present MZI-based linear block and saturable-absorber nonlinear model, a practical optical XOR formed by directly cascading NAND gates in the same way as electronic logic was not achieved without restoration.

Luo et al. [6] demonstrated that cascaded all-optical NAND-based logic is possible in a different platform based on passive diffractive optical networks, where logical values are encoded in the relative optical power of two spatially separated apertures, and the same diffractive NAND design was cascaded to realize AND, OR, NOT, and a half-adder. Their work shows that cascaded all-optical NAND-based logic is possible in another optical platform, but under a different signal encoding and routing scheme than the one used in the present silicon-photonics MZI-based study.

Future work can first investigate other silicon-photonics platforms and architectures, using different linear blocks and different nonlinear models or devices, to determine whether the cascading limitation observed here is specific to the present MZI-based ANN implementation or whether it remains for other silicon-photonics realizations of an optical NAND gate. A second future direction is to keep the same general platform used in this work, but give each logic stage its own trainable parameters or train the full cascaded circuit end-to-end rather than directly reusing a single trained NAND block.

VI. References

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VII. Code and simulations

[GitHub repository](#)

